

Monotone Treatment Response

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MONOTONE TREATMENT RESPONSE

BY CHARLES F. MANSKI¹

The standard formalization of the econometric analysis of treatment response assumes that each member of a population of interest receives one of a set of mutually exclusive and exhaustive treatments, and that the outcome under the realized treatment is observable. Outcomes under the nonrealized treatments are necessarily unobservable; hence these outcomes are censored. This paper investigates what may be learned about treatment response when it is assumed that response functions are monotone, semi-monotone, or concave-monotone. The analysis assumes nothing about the process of treatment selection and imposes no cross-individual restrictions on response. The basic idea is to determine, for every member of the population, the set of response functions that pass through that person's realized (treatment, outcome) pair and that are consistent with the functional-form assumption imposed. These person-specific findings are then explicitly aggregated across the population to determine what can be learned about the distribution of response. The findings have application to the econometric analysis of market demand and of production.

KEYWORDS: Identification, treatment responses.

1. INTRODUCTION

A LARGE LITERATURE ON THE ANALYSIS of treatment response assumes that each member j of a population J has some covariates $x_j \in X$ and an individual-specific response function $y_j(\cdot): T \rightarrow Y$ mapping the mutually exclusive and exhaustive treatments $t \in T$ into outcomes $y_j(t) \in Y$. Let $z_j \in T$ denote the treatment that j actually receives. Then $y_j \equiv y_j(z_j)$ is this individual's *realized outcome* and $y_j(t)$, $t \neq z_j$ are *latent outcomes*. It is standard to formalize the population as a measure space (J, Ω, P) of agents, with P a probability measure. Then $P[x, y(\cdot), z, y]$ gives the distribution of covariates, response functions, realized treatments, and realized outcomes.

Response functions are not observable but covariates, realized treatments, and realized outcomes may be observable. I assume that the analyst can empirically learn $P(x, z, y)$, say by observing (x, z, y) for a random sample of individuals. The analyst's problem is to combine this empirical evidence with prior information in order to learn about the distribution $P[y(\cdot)]$ of response functions.

What prior information might be assumed? The central maintained assumption of the subliterature analyzing experimental data is that assigned treatments

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are statistically independent of response functions.² The subliterate analyzing nonexperimental data exhibits enormous diversity in the prior information brought to bear. It has long been common to assume that z is statistically independent of $y(\cdot)$, conditional on the covariates x or some function thereof (see Rosenbaum and Rubin (1983)), but a sizable body of econometric research has been concerned with cases where it is implausible to assume that z is conditionally independent of $y(\cdot)$. Perhaps the earliest work of this type is the classical econometric analysis of linear simultaneous equations (see Hood and Koopmans (1953)), where it is assumed that the response functions $y_j(\cdot)$, $j \in J$, are all linear in t and have the same slope parameter. Another line of work imposes assumptions that restrict $P[y(\cdot), z|x]$ up to a finite dimensional parameter vector. Maddala (1983), Heckman and Robb (1985), and Manski (1995) review the econometric literature and give varying perspectives on it.

Whereas a dominant theme in the analysis of treatment response has been to impose assumptions strong enough to identify features of $P[y(\cdot)]$ of interest, the research program that I have carried forward explores what may be learned under weak assumptions (see Manski (1989, 1990, 1994, 1995, 1996, 1997)). The first step is a worst-case analysis imposing no assumptions at all on $P[x, y(\cdot), z, y]$. From this base, alternative weak assumptions are posed and their identifying power determined. My work is motivated by a concern for the credibility of reported findings about treatment response. The weaker are the assumptions imposed, the more credible are the findings. Concern for the wide credibility of findings also motivates the related work of Robins (1989), Balke and Pearl (1993), Imbens and Angrist (1994), Hotz, Mullins, and Sanders (1997), and Clements, Heckman, and Smith (1997).

This paper investigates what may be learned about treatment response when it is assumed that response functions are monotone, semi-monotone, or concave-monotone. Nothing is assumed about the process of treatment selection and no cross-individual restrictions on response are imposed. There are many applied settings in which a researcher may be confident that response is monotone, but be wary of assuming anything else. Among these are the economic analysis of demand and production, which I use throughout the paper to introduce ideas.

A body of work in economic theory is concerned with inference on demand and production functions under weak assumptions (see Varian (1982, 1984)), but differs in key respects from the present analysis. The work in economic theory supposes that treatments are not mutually exclusive and assumes that the researcher observes multiple realized (treatment, outcome) pairs for a given individual j . The problem of interest is to use these observations to learn about

²The classical analysis of randomized experiments, codified in Fisher (1935), assumes that there is perfect compliance with assigned treatments; hence z is statistically independent of $y(\cdot)$. Many modern studies are concerned with noncompliance, and so distinguish between an individual's realized treatment z and assigned treatment, say $d \in T$. Then randomization makes d statistically independent of $y(\cdot)$, but not necessarily z .

j 's response function $y_j(\cdot)$. The present analysis begins from the premise that treatments are mutually exclusive, so the researcher observes only one realized (treatment, outcome) pair for each individual. The objective is to learn about the population distribution of response functions, not about the response function of any one individual.

My starting point is a finding reported in Manski (1995, Section 6.4). Let T be an ordered set of treatments, let Y be the real line, and assume that $y_j(\cdot)$, $j \in J$, are weakly increasing functions on T . For any $r \in \mathbb{R}^1$ and $t \in T$, consider the upper tail probability $P[y(t) \geq r]$. I have shown that monotonicity of the response functions implies this sharp bound on $P[y(t) \geq r]$:³

$$(1) \quad P(t \geq z \cap y \geq r) \leq P[y(t) \geq r] \leq P(t > z \cup y \geq r).$$

The present analysis extends the simple argument used to prove (1) to derive sharp bounds on a variety of functionals of $P[y(\cdot)]$ that *respect stochastic dominance*. Let W be a collection of real random variables. Letting $v \in W$ and $w \in W$, v is said to stochastically dominate w if $P(v \geq r) \geq P(w \geq r)$ for all $r \in \mathbb{R}^1$. I say that a real functional $D(\cdot)$ on W respects stochastic dominance (or is a D -parameter) if $D(v) \geq D(w)$ whenever v stochastically dominates w . Familiar examples include upper tail probabilities, quantiles, and the means of weakly increasing functions of random variables. Measures of spread such as the variance and interquartile range are not D -parameters.⁴

Section 2 reports sharp bounds on three types of D -parameters under the assumption of monotone response. Proposition M1 gives the sharp bound on $D[y(t)]$, termed the D -outcome of treatment t . Propositions M2 and M3 give sharp bounds on two D -treatment effects, namely $D[y(t)] - D[y(s)]$ and $D[y(t) - y(s)]$, where $t \in T$ and $s \in T$ are any specified treatments. The two treatment effects coincide when D is the expectation functional but otherwise may differ. To distinguish them, I call $D[y(t)] - D[y(s)]$ a ΔD -treatment effect and $D[y(t) - y(s)]$ a $D\Delta$ -treatment effect.^{5,6} Section 2 concludes by interpreting the propositions in the framework of traditional econometric models of response to treatments and covariates.

³A new derivation of (1) is given in Section 2.2 below. The special case in which T contains two elements was examined in Manski (1994; 1995, Section 2.6) and the general case of ordered T in Manski (1995, Section 6.4). In these sources, monotone response was called *ordered outcomes*.

⁴I am aware of no accepted name for the class of parameters that respect stochastic dominance; hence I introduce the name " D -parameter" here. The phrase "respect stochastic dominance" was used in Horowitz and Manski (1995, Section 3.2), which studied the identification of such parameters under contaminated and corrupted sampling.

⁵Analyses of treatment effects seek to compare the outcomes that would occur if all $j \in J$ were to receive a specified treatment, say s , with the outcomes under an alternative specified treatment, say t . One way is to compare the outcome distributions $P[y(s)]$ and $P[y(t)]$. Another is to study the distribution $P[y(t) - y(s)]$ of the difference between the outcomes of the two treatments. These approaches are logically distinct, knowledge of $P[y(t) - y(s)]$ neither implying nor being implied by knowledge of $P[y(s)]$ and $P[y(t)]$. It is the case, however, that $E[y(t) - y(s)] = E[y(t)] - E[y(s)]$.

⁶When treatments are real-valued, $\{D[y(t)] - D[y(s)]\}/(t - s)$ is the ΔD -response slope and $D\{[y(t) - y(s)]/(t - s)\}$ is the $D\Delta$ -response slope.

Section 3 weakens the assumption of monotone response by supposing that the treatment set T is only semi-ordered rather than ordered. Whereas an ordered set orders every pair (s, t) of treatments, a semi-ordered set orders only a subset of these pairs. As in Section 2, I assume that $t \geq s \Rightarrow y_j(t) \geq y_j(s)$, all $j \in J$. When (s, t) is not ordered, however, the ordering of $y_j(t)$ and $y_j(s)$ is not known. So response is semi-monotone rather than monotone. Extension of Propositions M1 through M3 from monotone response to semi-monotone response is straightforward. These extensions are substantively important because they enlarge the domain of the analysis from problems with scalar variation in treatments to ones with vector variation.

Whereas Section 3 weakens the assumption of monotone response, Section 4 strengthens the assumption by imposing additional structure on response functions. I restrict attention to problems with nonnegative treatments and outcomes and I assume that response functions are concave-monotone. The findings have applications in economic production analysis, where it is often assumed that increasing a single input has *diminishing marginal returns*.

Section 5 gives conclusions.

2. MONOTONE RESPONSE

Section 2.1 uses the economic analysis of market demand to introduce ideas. Section 2.2 derives sharp bounds on D -outcomes. Section 2.3 derives sharp bounds on the two D -treatment effects. Section 2.4 interprets these treatment effects in terms of econometric models of response to treatments and covariates.

2.1. Downward Sloping Demand

I am concerned with situations in which an applied researcher is confident that response functions are monotone but is wary of imposing other assumptions. Economists analyzing market demand often face these conditions.

Economic analyses of market demand usually suppose there is a set of isolated markets for a given product. Each market is characterized by a demand function, which gives the quantity of product that price-taking consumers would purchase if price were set at any specified level. In each market, the interaction of consumers and firms determines the price at which transactions actually take place.

In the language of the present paper, markets are individuals, prices are treatments, and quantity demanded is an outcome. Thus T is the set of logically possible prices. In each market $j \in J$, transactions take place at some realized price $z_j \in T$. The market demand function is $y_j(\cdot)$ and $y_j \equiv y_j(z_j)$ is the quantity actually transacted in market j . The empirical evidence is composed of data on the covariates, prices, and quantities (x_i, z_i, y_i) , $i = 1, \dots, N$, realized in a random sample of N markets. The inferential problem is to combine this evidence with prior information to learn about the distribution $P[y(\cdot)]$ of demand functions across markets.

The one relatively firm conclusion of the economic theory of demand is that market demand ordinarily is a downward sloping function of price.⁷ The theory does not yield other conclusions about the shape of demand functions. Nor does the theory of demand imply anything about price determination. Conclusions about price determination can be drawn only if assumptions about the structure of demand are combined with assumptions about the behavior of the firms that produce the product in question. So demand analysis offers a good example of an inferential problem in which the analyst can reasonably assert that response functions are monotone but should be wary of imposing other assumptions.

It is intuitive that the assumption of downward sloping demand should have identifying power. Consider Figure 1. In the absence of prior information, observation of a realized (price, quantity) pair (z_j, y_j) reveals only that $y_j(\cdot)$ is some function passing through (z_j, y_j) . But if it is known that the demand function slopes downward, observation of (z_j, y_j) reveals that $y_j(\cdot)$ traverses the northwest (NW) and southeast (SE) regions of the figure.

Oddly, the classical econometric analysis of demand and supply as linear simultaneous equations does not assume that market demand is downward sloping. Instead, it imposes another assumption on the structure of demand functions. Begun in the 1920s, brought to maturity in Hood and Koopmans (1953), and expositied since in most every econometrics text, the classical analysis assumes that demand is a linear function of price, with the same slope param-

⁷This is not a universal prediction. Introductory textbooks expositing consumer theory distinguish between the substitution and income effects of price changes. When income effects are sufficiently strong, consumer optimization implies the existence of *Giffen goods*, for which demand increases with price over some domain. The modern theory of markets with imperfect information emphasizes that price may convey information. If the informational content of price is sufficiently strong, demand functions need not always slope downward. These exceptions notwithstanding, the ordinary presumption of economists is that demand functions are downward sloping.

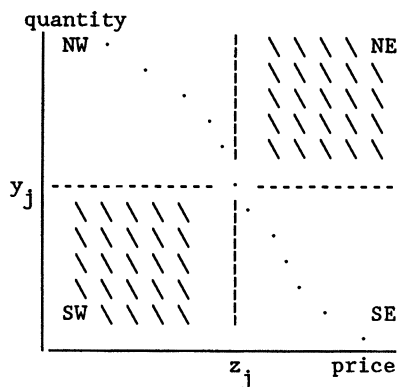


FIGURE 1.—Feasible demand functions.

ter in each market. Thus,

$$(2) \quad y_j(t) = \beta t + u_j,$$

where β is the common slope parameter and u_j is a market-specific intercept. Nothing is assumed about the sign or magnitude of β .

Economic theory does not suggest that demand should be linear in price and applied researchers rarely motivate the assumption. The main appeal of (2) is that it reduces the problem of inference on the distribution $P[y(\cdot)]$ of demand functions to one of inference on the scalar parameter β . The central classical finding is that $P[y(\cdot)]$ is identified if (2) is combined with this *exclusion restriction*:

$$(3a) \quad E(u|x=x_0) = E(u|x=x_1),$$

$$(3b) \quad E(z|x=x_0) \neq E(z|x=x_1),$$

where x_0 and x_1 are specified covariate values.⁸

2.2. Sharp Bounds on D-Outcomes

The formal analysis that begins here assumes that response functions are weakly increasing rather than decreasing. That is,

$$(4) \quad t \geq s \Rightarrow y_j(t) \geq y_j(s)$$

for all $j \in J$. With obvious modifications, the results apply to problems in which response functions are weakly decreasing, as in demand studies.⁹

In what follows, the outcome space Y is always a closed subset of the extended real line $[-\infty, \infty]$. The infimum and supremum of Y are denoted K_0

⁸Here is a simple proof taken from Manski (1995, page 152). Assumption (2) implies that $u_j = y_j - \beta z_j$ in each market j . This and (3a) imply that

$$(*) \quad E(y - \beta z|x=x_0) = E(y - \beta z|x=x_1).$$

Solving $(*)$ for β yields

$$(**) \quad \beta = \frac{E(y|x=x_0) - E(y|x=x_1)}{E(z|x=x_0) - E(z|x=x_1)},$$

provided that (3b) holds. Empirical observation of $P(x, z, y)$ identifies the conditional expectations $E(y|x)$ and $E(z|x)$ on the right side of $(**)$, so β is identified. Knowledge of β and $P(z, y)$ imply knowledge of $P(u)$, hence $P[y(\cdot)]$.

⁹In the economic analysis of competitive markets, it is often assumed that each realized (price, quantity) pair (z, y) is the intersection of two monotone response functions, an upward sloping market supply function and a downward sloping market demand function. Applied in this setting, our results assuming weakly increasing response concern the distribution of supply functions across markets. The parallel results assuming weakly decreasing response concern the distribution of demand functions across markets.

and K_1 respectively. The proof to Proposition M1 shows that

$$(5a) \quad y_{0j}(t) \equiv y_j \quad \text{if } t \geq z_j, \\ \equiv K_0 \quad \text{otherwise,}$$

$$(5b) \quad y_{1j}(t) \equiv y_j \quad \text{if } t \leq z_j, \\ \equiv K_1 \quad \text{otherwise,}$$

are sharp lower and upper bounds on $y_j(t)$. These bounds are central to the analysis.

With this background, Proposition M1 presents the sharp bound on D -outcomes implied by the assumption of monotone response. The proposition holds provided that the relevant D -parameter is well-defined.

PROPOSITION M1 (D -outcomes): *Let T be ordered. Let $y_j(\cdot)$, $j \in J$ be weakly increasing on T . Let Y be a closed subset of the extended real line. Let $D(\cdot)$ respect stochastic dominance. Then for every $t \in T$,*

$$(6) \quad D[y_0(t)] \leq D[y(t)] \leq D[y_1(t)].$$

This bound is sharp.

PROOF: Let $j \in J$. Monotonicity of $y_j(\cdot)$ implies this sharp bound on $y_j(t)$:

$$(7) \quad t < z_j \Rightarrow K_0 \leq y_j(t) \leq y_j, \\ t = z_j \Rightarrow y_j(t) = y_j, \\ t > z_j \Rightarrow y_j \leq y_j(t) \leq K_1.$$

Equivalently,

$$(8) \quad y_{0j}(t) \leq y_j(t) \leq y_{1j}(t).$$

There are no cross-individual restrictions, so the sharp joint bound on $[y_j(t), j \in J]$ is

$$(9) \quad y_{0j}(t) \leq y_j(t) \leq y_{1j}(t), \quad j \in J.$$

Hence $y_0(t)$ is stochastically dominated by $y(t)$, which in turn is stochastically dominated by $y_1(t)$. This shows that (6) is a bound on $D[y(t)]$.

The bound (6) is sharp because the bound (9) is sharp. That is, the empirical evidence and prior information are consistent with the hypothesis $\{y_j(t) = y_{0j}(t), j \in J\}$ and also with the hypothesis $\{y_j(t) = y_{1j}(t), j \in J\}$. *Q.E.D.*

Proposition M1 states what may be learned about D -outcomes using the assumption of monotone response and empirical knowledge of $P(z, y)$. Given data on the treatments and outcomes (z_i, y_i) , $i = 1, \dots, N$, realized by a random sample of N individuals, the bound (6) can be estimated by replacing $P(z, y)$

with the empirical distribution $P_N(z, y)$. For any $t \in T$, $D[y_0(t)]$ and $D[y_1(t)]$ can be estimated by the corresponding empirical D -statistics $D_N[y_0(t)]$ and $D_N[y_1(t)]$.

How informative is the bound? The answer depends on the D -parameter of interest and on the shape of $P(z, y)$. This is made clear in Corollaries M1.1 through M1.3, which apply Proposition M1 to upper tail probabilities, to the means of increasing functions, and to quantiles. In each case, the corollary is obtained by evaluating $D[y_0(t)]$ and $D[y_1(t)]$ for the D -parameter of interest.¹⁰

COROLLARY M1.1 (Upper Tail Probabilities): *Let $r \in (K_0, K_1]$. Then*

$$(10) \quad P(t \geq z \cap y \geq r) \leq P[y(t) \geq r] \leq P(t > z \cup y \geq r).$$

COROLLARY M1.2 (Means of Increasing Functions): *Let $f(\cdot): R^1 \rightarrow R^1$ be weakly increasing. Then*

$$(11) \quad \begin{aligned} f(K_0)P(t < z) + E[f(y)|t \geq z] \cdot P(t \geq z) \\ \leq E\{f[y(t)]\} \\ \leq f(K_1)P(t > z) + E[f(y)|t \leq z] \cdot P(t \leq z). \end{aligned}$$

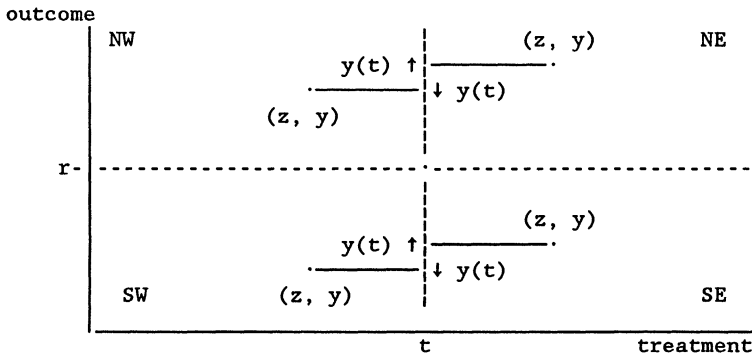
COROLLARY M1.3 (Quantiles): *Let $\alpha \in (0, 1)$. Let $q_\alpha(v)$ denote the α -quantile of a real random variable v . Let $\lambda_0 \equiv [\alpha - P(t < z)]/P(t \geq z)$ and $\lambda_1 \equiv \alpha/P(t \leq z)$. Then*

$$(12) \quad \begin{aligned} 0 < \alpha \leq P(t < z) &\Rightarrow K_0 \leq q_\alpha[y(t)] \leq q_{\lambda_1}(y|t \leq z), \\ P(t < z) < \alpha \leq P(t \leq z) &\Rightarrow q_{\lambda_0}(y|t \geq z) \leq q_\alpha[y(t)] \leq q_{\lambda_1}(y|t \leq z), \\ P(t \leq z) < \alpha < 1 &\Rightarrow q_{\lambda_0}(y|t \geq z) \leq q_\alpha[y(t)] \leq K_1. \end{aligned}$$

Figure 2 provides a simple way to interpret Corollary M1.1. The figure divides the realized (treatment, outcome) pairs into four regions based on the position of (z, y) relative to (t, r) . Each (z, y) pair in the NW region implies that $y(t) \geq r$. Hence $P[y(t) \geq r]$ is at least equal to the probability that (z, y) lies inside the NW region, namely $P(t \geq z \cap y \geq r)$. Each (z, y) pair in the SE region implies that $y(t) < r$. Hence $P[y(t) \geq r]$ is at most equal to the probability that (z, y) lies outside the SE region, namely $P(t > z \cup y \geq r)$. So we obtain the bound (10). The bound is sharp because (z, y) pairs in the SW or NE region do not reveal whether $y(t)$ is less than or greater than r .

The width of the bound varies with (t, r) and with the shape of $P(z, y)$. At one extreme, the mass of $P(z, y)$ may be concentrated entirely in the SW and NE regions of Figure 2. Then (10) becomes $0 \leq P[y(t) \geq r] \leq 1$, so the bound is uninformative. At the other extreme, the mass of $P(z, y)$ may be concentrated

¹⁰An alternative derivation of Corollary M1.1 and an empirical illustration are given in Manski (1995, Section 6.4).

FIGURE 2.—Identification of $P[y(t) \geq r]$.

entirely in the NW and SE regions. Then (10) becomes $P[y(t) \geq r] = P(t \geq z \cap y \geq r)$, so $P[y(t) \geq r]$ is identified. Thus some upper tail probabilities may remain completely unknown and others may be identified.

Upper tail probabilities are a subclass of the D -parameters covered by Corollary M1.2. The indicator function $1[y(t) \geq r]$ is weakly increasing in $y(t)$ and $E\{1[y(t) \geq r]\} = P[y(t) \geq r]$. It is revealing to compare Corollary M1.2, which exploits the assumption of monotone response, with the worst-case bound on $E\{f[y(t)]\}$ reported in Manski (1994).

The worst-case bound, which uses only the empirical evidence, is

$$\begin{aligned}
 (13) \quad & f(K_0)P(t \neq z) + E[f(y)|t = z] \cdot P(t = z) \\
 & \leq E\{f[y(t)]\} \\
 & \leq f(K_1)P(t \neq z) + E[f(y)|t = z] \cdot P(t = z).
 \end{aligned}$$

Whereas this bound draws information about $E\{f[y(t)]\}$ only from the (z, y) pairs with $t = z$, all (z, y) pairs are potentially informative under the monotone-response assumption. The lower bound in Corollary M1.2 draws information from the individuals with $t \geq z$ and the upper bound draws information from those with $t \leq z$. As shown in the discussion of upper tail probabilities, $E\{f[y(t)]\}$ may be identified even if $P(t = z) = 0$.

Corollary M1.3 shows that the monotone-response assumption generically yields a one-sided bound on quantiles of $y(t)$. The upper bound is informative when $\alpha \leq P(t \leq z)$. The lower bound is informative when $\alpha > P(t < z)$. These two cases exhaust the possibilities if $P(t = z) = 0$. The lower and the upper bounds are both informative if $P(t = z) > 0$ and $P(t < z) < \alpha \leq P(t \leq z)$.

2.3. Sharp Bounds on D -Treatment Effects

Propositions M2 and M3 present the sharp bounds on the D -treatment effects $D[y(t)] - D[y(s)]$ and $D[y(t) - y(s)]$.

PROPOSITION M2 (ΔD -treatment Effects): *Let T be ordered. Let $y_j(\cdot)$, $j \in J$ be weakly increasing on T . Let Y be a closed subset of the extended real line. Let $D(\cdot)$ respect stochastic dominance. Then for every $t \in T$ and $s \in T$ with $t > s$,*

$$(14) \quad 0 \leq D[y(t)] - D[y(s)] \leq D[y_1(t)] - D[y_0(s)].$$

This bound is sharp.

PROOF: Monotonicity of response implies that $y(t)$ stochastically dominates $y(s)$, so 0 is a lower bound on $D[y(t)] - D[y(s)]$. Proposition M1 implies that $D[y_1(t)] - D[y_0(s)]$ is an upper bound. We need to prove that these bounds are sharp.

Let $j \in J$. Monotonicity of $y_j(\cdot)$ implies this sharp joint bound on $y_j(t)$ and $y_j(s)$:

$$(15) \quad \begin{aligned} s < t < z_j &\Rightarrow K_0 \leq y_j(s) && \leq y_j(t) \leq y_j, \\ s < t = z_j &\Rightarrow K_0 \leq y_j(s) && \leq y_j(t) = y_j, \\ s < z_j < t &\Rightarrow K_0 \leq y_j(s) \leq y_j \leq y_j(t) \leq K_1, \\ s = z_j < t &\Rightarrow y_j = y_j(s) && \leq y_j(t) \leq K_1, \\ z_j < s < t &\Rightarrow y_j \leq y_j(s) && \leq y_j(t) \leq K_1. \end{aligned}$$

There are no cross-individual restrictions, so the empirical evidence and prior information are consistent with the hypothesis $\{y_j(s) = y_j(t), j \in J\}$ and also with the hypothesis $\{y_j(t) = y_{1j}(t), y_j(s) = y_{0j}(s), j \in J\}$. Hence (14) is sharp. *Q.E.D.*

PROPOSITION M3 ($D\Delta$ -treatment Effects): *Let T be ordered. Let $y_j(\cdot)$, $j \in J$ be weakly increasing on T . Let Y be a closed subset of the extended real line. Let $D(\cdot)$ respect stochastic dominance. Then for every $t \in T$ and $s \in T$ with $t > s$,*

$$(16) \quad D(0) \leq D[y(t) - y(s)] \leq D[y_1(t) - y_0(s)].$$

This bound is sharp.

PROOF: The proof to Proposition M2 showed that the sharp joint bound on $[y_j(t) - y_j(s), j \in J]$ is

$$(17) \quad 0 \leq y_j(t) - y_j(s) \leq y_{1j}(t) - y_{0j}(s), \quad j \in J.$$

Hence the degenerate distribution with all mass at 0 is stochastically dominated by $y(t) - y(s)$, which in turn is stochastically dominated by $y_1(t) - y_0(s)$. This shows that (16) is a bound on $D[y(t) - y(s)]$. The bound (16) is sharp because the bound (17) is sharp. *Q.E.D.*

Observe that the lower bounds in Propositions M2 and M3, namely 0 and $D(0)$, are implied by the monotone-response assumption and do not depend on the empirical evidence. The monotone-response assumption and the empirical evidence together determine the upper bounds.

Propositions M2 and M3 generically give distinct bounds for distinct treatment effects, but the two sets of bounds coincide when $D(\cdot)$ is the expectation functional. Applying Corollary M1.2, we have¹¹

COROLLARY M2.1 & M3.1 (Average Treatment Effects):

$$(18) \quad 0 \leq E[y(t)] - E[y(s)] = E[y(t) - y(s)] \\ \leq K_1 \cdot P(t > z) + E(y|t \leq z) \cdot P(t \leq z) \\ - K_0 \cdot P(s < z) - E(y|s \geq z) \cdot P(s \geq z).$$

The bound (18) takes a particularly simple form when outcomes are binary. Let Y be the two-element set $\{0, 1\}$. Then $K_0 = 0$, $K_1 = 1$, and (18) becomes

$$(19) \quad 0 \leq P[y(t) = 1] - P[y(s) = 1] = P[y(t) - y(s) = 1] \\ \leq P(y = 0, t > z) + P(y = 1, s < z).$$

2.4. *Econometric Models of Response to Treatments and Covariates*

Propositions M1 through M3 assume only that the response functions $y_j(\cdot)$, $j \in J$, are monotone. In other respects, the members of the population may have arbitrarily different response functions. I have found that the notation $y_j(\cdot)$ gives succinct and convenient expression to the idea that response functions may vary across the population (also see Angrist, Grady, and Imbens (1995)). Yet econometrics has a long tradition of expressing cross-individual variation in treatment response in terms of variation in covariates. This section interprets the propositions from that perspective.

Let each individual $j \in J$ have a covariate vector $u_j \in U$. These covariates may include the observable x discussed earlier, but there is no need here to distinguish observable from unobservable covariates. A standard econometric response model expresses $y_j(\cdot)$ as

$$(20) \quad y_j(t) = y^*(t, u_j).$$

The function $y^*(\cdot, \cdot)$ mapping $T \times U$ into Y is common to all $j \in J$.

In the econometric literature on discrete choice analysis initiated by McFadden (1973), for example, T is the choice set and $y^*(t, u_j)$ is the utility that an individual with covariates u_j associates with alternative t . An important way that discrete choice analysis differs from the analysis of treatment response is that the discrete choice analyst does not observe the realized utility outcome y_j . Only the identity z_j of the chosen alternative is observed.

In terms of (20), $y_j(t)$ is the outcome that individual j would experience if he were to receive treatment t , while holding his covariates fixed at the realized value u_j . Monotonicity of $y_j(\cdot)$ is equivalent to monotonicity of $y^*(\cdot, u_j)$, with

¹¹An alternative derivation of Corollary M2.1 is given in Pepper (1995).

$t \geq s \Rightarrow y^*(t, u_j) \geq y^*(s, u_j)$. The random variable $y(t)$ expresses the outcomes that would be experienced if all members of the population were to receive treatment t , while holding their covariates fixed at their realized values u_j , $j \in J$. The D -treatment effects $D[y(t)] - D[y(s)]$ and $D[y(t) - y(s)]$ compare the outcomes that would be experienced under treatments s and t if the covariates were held fixed at their realized values.

An alternative interpretation of $y_j(t)$ becomes available if we generalize the response model by supposing that variation in treatments induces variation in covariates. Let the *covariate response function* $u_j(\cdot): T \rightarrow U$ map treatments into covariates, let $u_j \equiv u_j(z_j)$, and replace (20) by

$$(20') \quad y_j(t) = y^*[t, u_j(t)].$$

In this formulation, $y_j(t)$ is the outcome that individual j would experience if he were to receive treatment t and his covariates were to take the value $u_j(t)$. Monotonicity of $y_j(\cdot)$ is equivalent to monotonicity of $y^*[\cdot, u_j(\cdot)]$ considered as a function of t . The D -treatment effects $D[y(t)] - D[y(s)]$ and $D[y(t) - y(s)]$ compare the outcomes experienced under treatments s and t taking account of induced variation in covariates.

The two interpretations of $y_j(t)$ are not contradictory. Propositions M1 through M3 apply to the thought experiment with covariates held fixed at realized values if $y^*(\cdot, u_j)$ is monotone on T . The propositions apply to the thought experiment with induced variation in covariates if $y^*[\cdot, u_j(\cdot)]$ is monotone on T . The propositions apply to both thought experiments if y^* is monotone in both senses. In this last case, one should not conclude that $y^*(t, u_j) = y^*[t, u_j(t)]$, but one can conclude that $y^*(t, u_j)$ and $y^*[t, u_j(t)]$ both lie within the common sharp bound $[y_{0j}(t), y_{1j}(t)]$.

From the perspective of econometric response models, there is no fundamental difference between treatments and covariates. I have designated t a treatment and u a covariate only because I have been performing thought experiments that vary t autonomously but not u . What if we want to perform thought experiments that autonomously vary some components of u ? Then we need to generalize the definition of T to be a set of treatment vectors, as in the next section.

3. SEMI-MONOTONE RESPONSE

In this section treatments are K -dimensional vectors and T is a semi-ordered set of treatment vectors. I write $s \not\geq t$ when a pair (s, t) is not ordered. Section 3.1 uses the economic analysis of production functions to introduce ideas. Section 3.2 presents the semi-monotone response versions of Proposition M1 and its corollaries. Section 3.3 presents the semi-monotone response versions of Propositions M2 and M3. Section 3.4 shows that the hypothesis of semi-monotone response is not testable but the joint hypothesis of semi-monotone response and exogenous treatment selection is testable.

3.1. *Production Analysis*

Economic analyses of production often suppose that firms or other entities use input vectors to produce a scalar output. Firm j has a production function $y_j(\cdot)$ mapping input vectors into product output, so $y_j(t)$ is the output that firm j produces when t is the input vector. The input vector z_j actually used by firm j and the firm's realized output $y_j(z_j)$ are determined by the firm's objective function and by the structure of the markets for inputs and outputs.

The most basic tenet of the economic theory of production is that output weakly increases with the level of the inputs. Let $s \equiv (s_1, s_2, \dots, s_K)$ and $t \equiv (t_1, t_2, \dots, t_K)$ be two input vectors, with $t_k \geq s_k$, all $k = 1, \dots, K$. Then $t \geq s$ and production theory predicts that $y_j(t) \geq y_j(s)$, $j \in J$. Suppose instead that there are input components k and k' such that $t_k > s_k$ and $t_{k'} < s_{k'}$. Then $s \not\leq t$ and production theory does not predict the ordering of $y_j(t)$ and $y_j(s)$. Thus production functions are semi-monotone.

To illustrate, consider the production of corn. The inputs include land and seed. The output is bushels of corn. Production theory predicts that the quantity of corn produced by a farm weakly increases with its input of land and seed. Production theory does not predict the effect on corn production of increasing one input component and decreasing the other. An analyst of corn production may want to learn a variety of treatment effects—the effect of changing the amount of seed planted on a fixed acreage of land, the effect of changing the acreage on which a fixed amount of seed is planted, or the effect of jointly changing the amount of seed and the acreage planted.

3.2. *Sharp Bounds on D-Outcomes*

The analysis of semi-monotone response uses much of the structure developed in Section 2 and so is amenable to succinct presentation. When T is semi-ordered, the definitions of $y_{0j}(t)$ and $y_{1j}(t)$ given in (5) remain valid, the term “otherwise” now including the possibility that $t \not\leq z_j$. Proposition S1 gives the semi-monotone response version of Proposition M1. Observe that the conclusion to Proposition M1 still holds and the proof requires only slight modification.

PROPOSITION S1 (D-outcomes): *Let T be semi-ordered. Let $y_j(\cdot)$, $j \in J$, be weakly increasing on the ordered pairs in T . Let Y be a closed subset of the extended real line. Let $D(\cdot)$ respect stochastic dominance. Then for every $t \in T$,*

$$(21) \quad D[y_0(t)] \leq D[y(t)] \leq D[y_1(t)].$$

This bound is sharp.

PROOF: Let $j \in J$. Semi-monotonicity of $y_j(\cdot)$ implies this sharp bound on $y_j(t)$:

$$(22) \quad \begin{aligned} t < z_j &\Rightarrow K_0 \leq y_j(t) \leq y_j, \\ t = z_j &\Rightarrow y_j(t) = y_j, \\ t > z_j &\Rightarrow y_j \leq y_j(t) \leq K_1, \\ t \neq z_j &\Rightarrow K_0 \leq y_j(t) \leq K_1. \end{aligned}$$

Thus (8) holds. The rest of the proof is the same as the proof to Proposition M1. *Q.E.D.*

Although Propositions M1 and S1 have the same stated conclusion, weakening the assumption of monotone response to semi-monotone response is consequential. Suppose that an ordering of the treatment set T is weakened to a semi-ordering. Each time that a pair (t, z_j) with $t > z_j$ becomes unordered, $y_{0j}(t)$ falls from y_j to K_0 . Each time that a pair (t, z_j) with $t < z_j$ becomes unordered, $y_{1j}(t)$ rises from y_j to K_1 . Hence the ordered- T version of $y_0(t)$ stochastically dominates its semi-ordered- T counterpart and the ordered- T version of $y_1(t)$ is stochastically dominated by its semi-ordered- T counterpart. So weakening the assumption of monotone response to semi-monotone response widens the bound on $D[y(t)]$. In the extreme case where T is entirely unordered, Proposition S1 gives the worst-case bound on $D[y(t)]$.

Corollaries S1.1 through S1.3 give the semi-monotone response versions of Corollaries M1.1 through M1.3. The explicit forms for $D[y_0(t)]$ and $D[y_1(t)]$ given in these corollaries show clearly how weakening the assumption of monotone response to semi-monotone response affects the bounds on D -outcomes.

COROLLARY S1.1 (Upper Tail Probabilities): *Let $r \in (K_0, K_1]$. Then*

$$(23) \quad P(t \geq z \cap y \geq r) \leq P[y(t) \geq r] \leq P(t > z \cup t \neq z \cup y \geq r).$$

COROLLARY S1.2 (Means of Increasing Functions): *Let $f(\cdot): R^1 \rightarrow R^1$ be weakly increasing. Then*

$$(24) \quad \begin{aligned} f(K_0)P(t < z \cup t \neq z) + E[f(y)|t \geq z] \cdot P(t \geq z) \\ \leq E\{f[y(t)]\} \\ \leq f(K_1)P(t > z \cup t \neq z) + E[f(y)|t \leq z] \cdot P(t \leq z). \end{aligned}$$

COROLLARY S1.3 (Quantiles): *Let $\alpha \in (0, 1)$. Let $q_\alpha(v)$ denote the α -quantile of a real random variable v . Let $\lambda_0 \equiv [\alpha - P(t < z \cup t \neq z)]/P(t \geq z)$ and $\lambda_1 \equiv$*

$\alpha/P(t \leq z)$. Then

$$\begin{aligned}
 (25) \quad & 0 < \alpha \leq P(t < z \cup t \neq z) \Rightarrow K_0 \leq q_\alpha[y(t)], \\
 & P(t < z \cup t \neq z) < \alpha < 1 \Rightarrow q_{\lambda_0}(y|t \geq z) \leq q_\alpha[y(t)], \\
 & 0 < \alpha \leq P(t \leq z) \Rightarrow q_\alpha[y(t)] \leq q_{\lambda_1}(y|t \leq z), \\
 & P(t \leq z) < \alpha < 1 \Rightarrow q_\alpha[y(t)] \leq K_1.
 \end{aligned}$$

3.3. Sharp Bounds on D-Treatment Effects

When T is semi-ordered, the conclusions to Propositions M2 and M3 still hold if $t < s$ and the upper bounds still hold if $t \neq s$. The lower bounds, however, need to be modified if $t \neq s$. Propositions S2 and S3 give these extensions to the earlier results.

PROPOSITION S2 (ΔD -treatment Effects): Let T be semi-ordered. Let $t \in T$ and $s \in T$. Let $y_j(\cdot)$, $j \in J$, be weakly increasing on the ordered pairs in T . Let Y be a closed subset of the extended real line. Let $D(\cdot)$ respect stochastic dominance. For $t > s$, the sharp bound on $D[y(t)] - D[y(s)]$ is

$$(26) \quad 0 \leq D[y(t)] - D[y(s)] \leq D[y_1(t)] - D[y_0(s)].$$

For $t \neq s$, the sharp bound is

$$(27) \quad D[y_0(t)] - D[y_1(s)] \leq D[y(t)] - D[y(s)] \leq D[y_1(t)] - D[y_0(s)].$$

PROOF: Let $t > s$. Semi-monotonicity of response implies that $y(t)$ stochastically dominates $y(s)$, so 0 is a lower bound on $D[y(t)] - D[y(s)]$. Proposition S1 implies that $D[y_1(t)] - D[y_0(s)]$ is an upper bound. To prove that these bounds are sharp, consider $j \in J$. If s , t , and z_j are ordered, (15) still gives the sharp joint bound on $y_j(t)$ and $y_j(s)$. If $z_j \neq t$ and/or $z_j \neq s$, the sharp joint bound on $y_j(t)$ and $y_j(s)$ is

$$\begin{aligned}
 (28) \quad & s < t \cap s < z_j \Rightarrow K_0 \leq y_j(s) \leq y_j \cap y_j(s) \leq y_j(t) \leq K_1, \\
 & s < t \cap z_j < t \Rightarrow K_0 \leq y_j(s) \leq y_j(t) \cap y_j \leq y_j(t) \leq K_1, \\
 & s < t \Rightarrow K_0 \leq y_j(s) \leq y_j(t) \leq K_1.
 \end{aligned}$$

The rest of the proof is the same as the proof to Proposition M2.

Let $s \neq t$. Proposition S1 implies that (27) is a bound on $D[y(t)] - D[y(s)]$. For each $j \in J$, the sharp joint bound on $y_j(t)$ and $y_j(s)$ is

$$\begin{aligned}
 (29) \quad & y_{0j}(t) \leq y_j(t) \leq y_{1j}(t), \\
 & y_{0j}(s) \leq y_j(s) \leq y_{1j}(s).
 \end{aligned}$$

There are no cross-individual restrictions, so the empirical evidence and prior information are consistent with the hypothesis $\{y_j(t) = y_{0j}(t) \text{ and } y_j(s) = y_{1j}(s)\}$,

$j \in J$ and also with the hypothesis $\{y_j(t) = y_{1j}(t) \text{ and } y_j(s) = y_{0j}(s), j \in J\}$. Hence (27) is sharp. *Q.E.D.*

PROPOSITION S3 (DA-treatment Effects): *Let T be semi-ordered. Let $t \in T$ and $s \in T$. Let $y_j(\cdot)$, $j \in J$, be weakly increasing on the ordered pairs in T . Let Y be a closed subset of the extended real line. Let $D(\cdot)$ respect stochastic dominance. For $t > s$, then the sharp bound on $D[y(t) - y(s)]$ is*

$$(30) \quad D(0) \leq D[y(t) - y(s)] \leq D[y_1(t) - y_0(s)].$$

For $s \neq t$, the sharp bound is

$$(31) \quad D[y_0(t) - y_1(s)] \leq D[y(t) - y(s)] \leq D[y_1(t) - y_0(s)].$$

PROOF: Let $t > s$. By (15) and (28), the sharp joint bound on $[y_j(t) - y_j(s), j \in J]$ is

$$(32) \quad 0 \leq y_j(t) - y_j(s) \leq y_{1j}(t) - y_{0j}(s), \quad j \in J.$$

The rest of the proof is the same as the proof to Proposition M3.

Let $s \neq t$. By (29), the sharp joint bound on $[y_j(t) - y_j(s), j \in J]$ is

$$(33) \quad y_{0j}(t) - y_{1j}(s) \leq y_j(t) - y_j(s) \leq y_{1j}(t) - y_{0j}(s), \quad j \in J.$$

Hence (31) is the sharp bound on $D[y(t) - y(s)]$. *Q.E.D.*

3.4. Testing the Hypothesis of Semi-Monotone Response

Propositions S1 through S3 assume that response is semi-monotone and deduce the implications for the analysis of treatment response. It is easy to see that the hypothesis of semi-monotone response is not testable in isolation. For each $j \in J$ only one point on the response function $y_j(\cdot)$ is observable, namely $y_j(z_j)$. Hence the empirical evidence is necessarily consistent with the hypothesis that $y_j(\cdot)$ is weakly increasing on the ordered pairs in T . In particular, the empirical evidence is consistent with the hypothesis that every response function is flat, with $\{y_j(t) = y_j, t \in T, j \in J\}$. So hypotheses that treatment effects are zero are not testable.

A researcher wishing to test the hypothesis of semi-monotone response can do so only if this hypothesis is joined with other assumptions. Consider in particular the common assumption that z is statistically independent of $y(\cdot)$; that is,

$$(34) \quad p[y(\cdot)] = P[y(\cdot)|z].$$

This assumption, often termed *exogenous treatment selection*, is not testable in isolation (see Manski (1995, Chapter 2)). However the joint hypothesis of semi-monotone response and exogenous treatment selection is testable.

The key is Proposition S4.

PROPOSITION S4: Let T be semi-ordered. Let $t \geq s$. Let $y_j(\cdot)$, $j \in J$, be weakly increasing on the ordered pairs in T . Let Y be a closed subset of the extended real line. Let z be statistically independent of $y(\cdot)$. Then $P(y|z=t)$ stochastically dominates $P(y|z=s)$.

PROOF: Semi-monotonicity implies that $y(t)$ stochastically dominates $y(s)$. Exogenous treatment selection implies that $P[y(s)] = P(y|z=s)$ and that $P[y(t)] = P(y|z=t)$. Q.E.D.

Empirical knowledge of the distribution $P(z, y)$ of realized treatments and outcomes implies knowledge of $P(y|z=s)$ and $P(y|z=t)$ for s and t on the support of $P(z)$. So Proposition 4 implies this test:

Reject the hypothesis of semi-monotone response and exogenous treatment selection if there are $s \in T$ and $t \in T$ on the support of $P(z)$ such that $t \geq s$ but $P(y|z=t)$ does not stochastically dominate $P(y|z=s)$.

In finite-sample practice, a researcher observing a random sample of (z, y) pairs can estimate $P(y|z=t)$ and $P(y|z=s)$ and form an asymptotically valid version of the test (see McFadden (1989)).

There are three ways to interpret an empirical finding that $P(y|z=t)$ does not stochastically dominate $P(y|z=s)$. Researchers confident that treatment selection is exogenous would conclude that response is not semi-monotone. Researchers confident that response is semi-monotone would conclude that treatment selection is not exogenous. Other researchers would conclude only that some part of the joint hypothesis is incorrect.

4. CONCAVE-MONOTONE RESPONSE

Whereas Section 3 weakened the assumption of monotone response, I now strengthen the assumption by imposing additional structure on response functions. In particular, I suppose that $y_j(\cdot)$, $j \in J$, are concave-monotone functions mapping nonnegative treatments into nonnegative outcomes.

Section 4.1 uses the economic analysis of production functions to introduce ideas. Section 4.2 derives sharp bounds on D -outcomes. Section 4.3 derives bounds on the two D -treatment effects. With appropriate modifications, the present analysis can be redone to yield bounds applicable when response functions are convex and monotone.

4.1. Short Run Production Analysis with Diminishing Marginal Returns

Consider again the economic analysis of production. In applications, researchers usually assume more than that production functions are semi-monotone. It is generally assumed that inputs and outputs are nonnegative. It is also common to assume *diminishing marginal returns*, which means that the production function is concave in each input component, holding the other components fixed.

In so-called short-run production analysis, researchers distinguish two types of inputs—*variable inputs* whose values can be changed and *fixed inputs* whose values cannot be changed. Short-run production analysis performs thought experiments in which the variable inputs are varied and the fixed inputs are held fixed at their realized values. Thus the variable inputs are considered to be treatments, the fixed inputs to be covariates, and the short-run production function maps variable inputs into output. Suppose that there is one variable input. Then it is common to assume that the short-run production function $y_j(\cdot)$ is concave-monotone in this input.

In the short-run production of corn, for example, seed would usually be thought of as a variable input and land as a fixed input. A researcher might find it plausible to assume that, holding land fixed, output of corn rises with the input of seed but with diminishing returns.

Figure 3 visually shows the identifying power of assuming that production functions are concave-monotone with nonnegative inputs and outputs. If response is assumed to be monotone, observation of a realized (input, output) pair (z_j, y_j) reveals that $y_j(\cdot)$ traverses the southwest (SW) and northeast (NE) regions of the figure. If it is additionally assumed that response is concave, we can conclude that $y_j(\cdot)$ traverses the west southwest (WSW) and east northeast (ENE) regions of the figure. The ray from the origin with slope y_j/z_j becomes the lower bound on $y_j(t)$ for $t \leq z_j$ and the upper bound on $y_j(t)$ for $t \geq z_j$.

An intrepid researcher willing to assume more than concavity and monotonicity may be able to conclude even more about the structure of production functions. Suppose, for example, that one is willing to go so far as to assume that the short run production function is linear in t and passes through the origin. Then observation of (z_j, y_j) identifies $y_j(\cdot)$, with $y_j(t) = (y_j/z_j)t$.

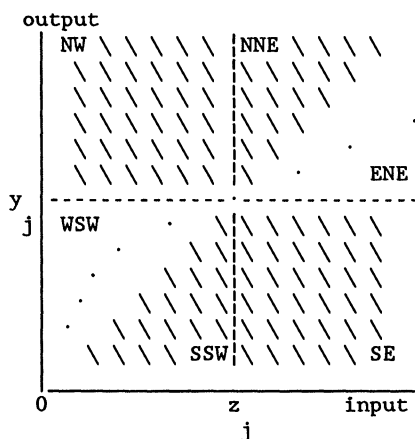


FIGURE 3.—Feasible concave monotone production functions.

4.2. Sharp Bounds on D-Outcomes

Here and in Section 4.3, $T = [0, \tau]$ for some $\tau \in (0, \infty]$ and $Y = [0, \infty]$. The important feature of this specification of treatments and outcomes is that T and Y have known finite lower bounds. Without this condition, assuming that response is concave-monotone has no identifying power beyond assuming that response is monotone.¹² Specifying that the lower bounds of T and Y are zero and that the upper bound of Y is ∞ merely permits some simplification in the analysis.

I repeatedly use this fact: Given three points $(v_m, w_m) \in [0, \infty]^2$, $m = 1, 2, 3$, with $0 < v_1 < v_2 < v_3$, there exists a concave-monotone function mapping $[0, \tau] \rightarrow [0, \infty]$ and passing through the three points if and only if

$$(35) \quad w_1/v_1 \geq (w_2 - w_1)/(v_2 - v_1) \geq (w_3 - w_2)/(v_3 - v_2) \geq 0.$$

Here w_1/v_1 is the slope of the line segment connecting the origin to (v_1, w_1) and $(w_m - w_{m-1})/(v_m - v_{m-1})$ is the slope of the line segment connecting (v_{m-1}, w_{m-1}) to (v_m, w_m) , $m = 2, 3$. In particular, the piecewise linear function passing through the origin and the three points is concave-monotone if and only if (35) holds.

Whereas $y_{0j}(t)$ and $y_{1j}(t)$ are sharp bounds on $y_j(t)$ when response is monotone, the proof to Proposition C1 shows that

$$(36a) \quad y_{c0j}(t) \equiv y_j \quad \text{if } t \geq z_j, \\ \equiv y_j t / z_j \quad \text{otherwise,}$$

$$(36b) \quad y_{c1j}(t) \equiv y_j \quad \text{if } t \leq z_j, \\ \equiv y_j t / z_j \quad \text{otherwise,}$$

¹²Even assuming that response is linear-monotone has no additional identifying power. Let $Y = [-\infty, \infty]$ and assume that

$$y_j(t) = \beta_j t + u_j,$$

where $\beta_j \geq 0$. Observation of (z_j, y_j) reveals that $u_j = y_j - \beta_j z_j$, so

$$y_j(t) = \beta_j(t - z_j) + y_j.$$

For $s \in T$ and $t \in T$ with $t > s$, the sharp joint bound on $y_j(s)$ and $y_j(t)$ is

$$s < t < z_j \Rightarrow -\infty \leq y_j(s) \leq y_j(t) \leq y_j,$$

$$s < t = z_j \Rightarrow -\infty \leq y_j(s) \leq y_j(t) = y_j,$$

$$s < z_j < t \Rightarrow -\infty \leq y_j(s) \leq y_j \leq y_j(t) \leq \infty,$$

$$s = z_j < t \Rightarrow y_j = y_j(s) \leq y_j(t) \leq \infty,$$

$$z_j < s < t \Rightarrow y_j \leq y_j(s) \leq y_j(t) \leq \infty.$$

This is the same as the bound (15) obtained when it was assumed only that response is monotone. It follows that adding the linearity assumption leaves unchanged the conclusions to Propositions M1 through M3.

are the sharp bounds when response is concave-monotone. Proposition C1 uses these bounds to find the sharp bound on D -outcomes.

PROPOSITION C1 (D -outcomes): Let $T = [0, \tau]$. Let $y_j(\cdot)$, $j \in J$, be concave and weakly increasing on T . Let $Y = [0, \infty]$. Let $D(\cdot)$ respect stochastic dominance. Then for every $t \in T$,

$$(37) \quad D[y_{c0}(t)] \leq D[y(t)] \leq D[y_{c1}(t)].$$

This bound is sharp.

PROOF: Let $j \in J$. Application of (35) yields this sharp bound on $y_j(t)$:

$$(38) \quad \begin{aligned} t < z_j &\Rightarrow y_j t / z_j \leq y_j(t) \leq y_j, \\ t = z_j &\Rightarrow y_j(t) = y_j, \\ t > z_j &\Rightarrow y_j(t) \leq y_j t / z_j. \end{aligned}$$

Equivalently,

$$(39) \quad y_{c0j}(t) \leq y_j(t) \leq y_{c1j}(t).$$

The rest of the proof is the same as the proof to Proposition M1, with $y_{c0j}(t)$ and $y_{c1j}(t)$ replacing $y_{0j}(t)$ and $y_{1j}(t)$. *Q.E.D.*

Comparison of the intervals $[y_{c0j}(t), y_{c1j}(t)]$ and $[y_{0j}(t), y_{1j}(t)]$ indicates that strengthening the assumption of monotone response to concave-monotone response has substantial identifying power. The present bound on $y_j(t)$ is not only narrower than the earlier one but its width varies with t in a qualitatively different manner. The present bound has width $y_j \cdot |(z_j - t)/z_j|$, whereas the earlier bound has width $y_j \cdot 1[t < z_j] + \infty \cdot 1[t > z_j]$. Thus the present bound widens linearly from zero as t moves away from z_j , whereas the width of the earlier bound varies discontinuously with t .

Corollaries C1.1 through C1.3 give the concave-monotone response versions of Corollaries M1.1 through M1.3. Comparison of these corollaries with the earlier ones shows clearly the additional identifying power of assuming that response is concave. The earlier bounds on $E\{f[y(t)]\}$ are uninformative if $f(K_0) = -\infty$ and $f(K_1) = \infty$, but the present bounds are essentially always informative. The earlier bounds on quantiles of $y(t)$ are generically informative only from one side or the other, but the present bounds are informative both from above and below.

COROLLARY C1.1 (Upper Tail Probabilities): Let $r \in (0, \infty]$. Then

$$(40) \quad \begin{aligned} P[(t \geq z \cap y \geq r) \cup (t < z \cap yt/z \geq r)] \\ \leq P[y(t) \geq r] \\ \leq P[(t > z \cap yt/z \geq r) \cup (t \leq z \cap y \geq r)]. \end{aligned}$$

COROLLARY C1.2 (Means of Increasing Functions): Let $f(\cdot): R^1 \rightarrow R^1$ be weakly increasing. Then

$$(41) \quad \begin{aligned} E[f(yt/z)|t < z] \cdot P(t < z) + E[f(y)|t \geq z] \cdot P(t \geq z) \\ \leq E\{f[y(t)]\} \\ \leq E[f(yt/z)|t > z] \cdot P(t > z) + E[f(y)|t \leq z] \cdot P(t \leq z). \end{aligned}$$

COROLLARY C1.3 (Quantiles): Let $\alpha \in (0, 1)$. Let $q_\alpha(v)$ denote the α -quantile of a real random variable v . Then

$$(42) \quad \begin{aligned} q_\alpha\{y \cdot 1[t \geq z] + yt/z \cdot 1[t < z]\} \leq q_\alpha[y(t)] \\ \leq q_\alpha\{y \cdot 1[t \leq z] + yt/z \cdot 1[t > z]\}. \end{aligned}$$

4.3. Bounds on D-Treatment Effects

Bounds on D -treatment effects follow from the sharp joint bounds obtained for $y_j(t)$ and $y_j(s)$. In Sections 2.3 and 3.3, we found these joint bounds to have simple forms when response is monotone or semi-monotone. The joint bounds assuming concave-monotone response are more complex. Application of (35) yields these joint restrictions on $y_j(t)$ and $y_j(s)$:

$$(43) \quad \begin{aligned} s < t < z_j \Rightarrow y_j(s)/s &\geq [y_j(t) - y_j(s)]/(t - s) \\ &\geq [y_j - y_j(t)]/(z_j - t) \geq 0, \\ s < t = z_j \Rightarrow y_j(s)/s &\geq [y_j(t) - y_j(s)]/(t - s) \\ &= [y_j - y_j(s)]/(z_j - s) \geq 0, \\ s < z_j < t \Rightarrow y_j(s)/s &\geq [y_j - y_j(s)]/(z_j - s) \\ &\geq [y_j(t) - y_j]/(t - z_j) \geq 0, \\ s = z_j < t \Rightarrow y_j(s)/s = y_j/z_j &\geq [y_j(t) - y_j]/(t - z_j) \geq 0, \\ z_j < s < t \Rightarrow y_j(s)/z_j &\geq [y_j(s) - y_j]/(s - z_j) \\ &\geq [y_j(t) - y_j(s)]/(t - s) \geq 0. \end{aligned}$$

Proposition C3 uses (43) to derive the sharp bound on $D[y(t) - y(s)]$. Proposition C2 gives the sharp lower bound on $D[y(t)] - D[y(s)]$ but only a nonsharp upper bound. Although Proposition C2 does not provide the sharp upper bound on general ΔD -treatment effects, application of Proposition C3 does yield the sharp upper bound on the average treatment effect $E[y(t)] - E[y(s)]$.

PROPOSITION C2 (ΔD -treatment Effects): Let $T = [0, \tau]$. Let $y_j(\cdot)$, $j \in J$, be concave and weakly increasing on T . Let $Y = [0, \infty]$. Let $D(\cdot)$ respect stochastic dominance. Then for every $t \in T$ and $s \in T$ with $t > s$,

$$(44) \quad 0 \leq D[y(t)] - D[y(s)] \leq D[y_{c1}(t)] - D[y_{c0}(s)].$$

The lower bound is sharp but the upper bound is not sharp.

PROOF: Monotonicity of response implies that $y(t)$ stochastically dominates $y(s)$, so 0 is a lower bound on $D[y(t)] - D[y(s)]$. This lower bound is sharp because the hypothesis $\{y_j(t) = y_j(s) = y_j, j \in J\}$ satisfies (43).

Proposition C1 implies that $D[y_{c1}(t)] - D[y_{c0}(s)]$ is an upper bound on $D[y(t)] - D[y(s)]$. This upper bound is not sharp because the hypothesis $\{y_j(t) = y_{c1j}(t), y_j(s) = y_{c0j}(s), j \in J\}$ does not satisfy (43). When $s < t < z_j$, setting $\{y_j(t) = y_j, y_j(s) = y_j s / z_j\}$ violates (43). Similarly, when $z_j < s < t$, setting $\{y_j(t) = y_j t / z_j, y_j(s) = y_j\}$ violates (43). Q.E.D.

PROPOSITION C3 ($D\Delta$ -treatment Effects): Let $T = [0, \tau]$. Let $y_j(\cdot)$, $j \in J$, be concave and weakly increasing on T . Let $Y = [0, \infty]$. Let $D(\cdot)$ respect stochastic dominance. For each $t \in T$ and $s \in T$ with $t > s$, define

$$(45) \quad y_{ctj}(s) \equiv y_j s / t \quad \text{if } t < z_j, \\ \equiv y_j s / z_j \quad \text{otherwise.}$$

Then

$$(46) \quad D(0) \leq D[y(t) - y(s)] \leq D[y_{c1}(t) - y_{ct}(s)].$$

This bound is sharp.

PROOF: The lower bound holds because response is monotone. It is sharp because the hypothesis $\{y_j(t) - y_j(s) = 0, j \in J\}$ satisfies (43).

To obtain the sharp upper bound, we need to determine the largest value of $y_j(t) - y_j(s)$ that satisfies (43). This can be accomplished in two steps. First hold $y_j(t)$ fixed and minimize $y_j(s)$ subject to (43). This yields the maximum of $y_j(t) - y_j(s)$ as a function of $y_j(t)$. Then maximize this expression over $y_j(t) \in [y_{c0j}(t), y_{c1j}(t)]$.

When $t < z_j$, setting $\{y_j(t) = y_j, y_j(s) = y_j s / t\}$ yields the maximal value of $y_j(t) - y_j(s)$. When $t \geq z_j$, setting $\{y_j(t) = y_j t / z_j, y_j(s) = y_j s / z_j\}$ yields the maximal value of $y_j(t) - y_j(s)$. It follows that the sharp joint bound on $[y_j(t) - y_j(s), j \in J]$ is

$$(47) \quad 0 \leq y_j(t) - y_j(s) \leq y_{c1j}(t) - y_{ctj}(s), \quad j \in J.$$

Hence $D[y_{c1}(t) - y_{ct}(s)]$ is the sharp upper bound on $D[y(t) - y(s)]$. Q.E.D.

Let us evaluate the bound in Proposition C3 for the expectation functional. Writing $y_{c1}(t) - y_{c1}(s)$ in the explicit form

$$(48) \quad y_{c1}(t) - y_{c1}(s) = 1[t < z] \cdot (t - s) \cdot y/t + 1[t \geq z] \cdot (t - s) \cdot y/z,$$

we obtain the following corollary.

COROLLARY C2.1 & C3.1 (Average Treatment Effects):

$$(49) \quad 0 \leq E[y(t)] - E[y(s)] = E[y(t) - y(s)] \\ \leq (t - s) \cdot [E(y/t | t < z) \cdot P(t < z) + E(y/z | t \geq z) \cdot P(t \geq z)].$$

5. CONCLUSION

Propositions 1 through 3 have established what may be learned about D -outcomes, ΔD -treatment effects, and $D\Delta$ -treatment effects when response functions are assumed to be monotone, semi-monotone, or concave-monotone. Corollaries apply the propositions to upper tail probabilities, means of increasing functions, and quantiles. The propositions and corollaries assume nothing about the process of treatment selection and impose no cross-individual restrictions on response functions.

The economics of demand and production should suffice to motivate applied interest in the findings, but it is easy to suggest other applications. Consider, for example, the problem of learning the health effects of cigarette smoking from data on (smoking, health) realizations. Medical science provides a firm basis for assuming that various health outcomes decline as an individual's smoking intensity rises. Neither medical nor behavioral science, however, offers a persuasive explanation of treatment selection; why some individuals smoke a lot and others not at all.

Perhaps the central theoretical conclusion that a reader should take away from this paper is that the assumptions of monotone response and of concave-monotone response qualitatively improve the identification problem relative to the worst-case situation in which no prior information is available. Using the empirical evidence alone, the only observations of (z, y) that are informative about outcomes at treatment t are those in which $z = t$. Using the empirical evidence and the monotone-response assumption, every observation yields either an informative lower or upper bound on outcomes at treatment t . Using the empirical evidence and the assumption of concave-monotone response, every observation yields an informative lower and upper bound on outcomes at t .

Of course the assumption of monotone response does not suffice to answer all questions of interest about treatment response. For example, it does not suffice to test commonly posed hypotheses that treatment effects are zero. A researcher wanting to learn more about treatment response must bring to bear other assumptions about the structure of response functions, impose cross-individual restrictions on response, and/or introduce assumptions about the process of treatment selection.

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